Team Number: ______

PROBLEM 1
10% of total

TOTAL	A	B	C	D	E	F
16	**2**	**2**	**3**	**3**	**4**	**2**

General Grading Scheme: up to ½ credit if final answer incorrent, -1 pt if no units

A strip of magnesium ribbon is dropped into hot water, and the following reaction takes place:

1 $Mg(s)$ + 2 $H_2O(l)$ → 1 $Mg(OH)_2(s)$ + 1 $H_2(g)$ 2 pts

a. Balance the reaction by filling in the coefficients, using lowest whole numbers.

b. Why must the magnesium ribbon be dropped into hot water to induce a reaction?

1 pt Mg reacts slowly with cold water

1 pt Hot water makes reaction faster

c. After a period of time, the reaction slows and the evolution of bubbles ceases. Explain these observations. Hint: Think about solubility.

1 pt coat of $Mg(OH)_2$ forms

1 pt $Mg(OH)_2$ is insoluble

1 pt H_2 production ceases with no more reaction

d. Assuming that the magnesium reacts completely with the water, calculate the mass of H_2 formed when 0.420g of magnesium ribbon is introduced.

Mass of H_2 = 0.0346g H_2 (3 pts)

e. The hydrogen gas evolved is captured and combusted with 0.250g of oxygen. If 0.230g of water are produced, find the limiting reagent and calculate the % yield.

Limiting Reagent: O_2 (2pts) *% Yield = 81.8% (2 pts)*

f. When magnesium is reacted with hot steam, it reacts with it in a 1:1 ratio to form two products. Write the equation of the balanced reaction.

Balanced Equation: $Mg(s) + H_2O(g) \rightarrow MgO(s) + H_2(g)$ 2 pts for correct products

Team Number: ______

PROBLEM 2
11% of total

TOTAL	A	B	C	D	E	F
18	**5**	**3**	**2**	**2**	**4**	**2**

An unknown metal X from the Group 1 elements (alkali metals) burns in a pure oxygen environment with an intense red flame to form a white compound.

a. Write the product of combustion (the metal oxide) that predominates for each the alkali metals, from lithium to cesium.

Li: Li_2O
Na: Na_2O_2
K: KO_2 and K_2O_2 both accepted
Rb: RbO_2
Cs: CsO_2 (1 pt each)

b. Assuming that 0.200g of the unknown metal X forms 0.431g of the oxide, identify X.

Identity of X: Li (3 pts)

c. When X is burned in atmospheric air, the metal oxide is not the exclusive product. The metal also reacts with nitrogen to form a metal nitride product. Write down the balanced equation of this nitrogen reaction (use X in place of the metal if you could not determine its identity).

Balanced Equation: $6\ Li(s) + N_2(s) \rightarrow 2\ Li_3N\ (s)$ (2 pts for product, -1 if not balanced)

d. The metal nitride reacts with water to form a gaseous product. Identify this gas.

Identity of Gas: NH_3 (2 pts)

e. 0.300g of metal X are burned in atmospheric air to form 0.600g of a mixture of the metal oxide and nitride. Find the mass of each product formed.

Mass of Metal Oxide = 0.442g *Mass of Metal Nitride = 0.158g*
(2 pts each)

f. Metal X is treated with hydrogen gas to form an ionic compound. Identify this ionic compound.

Identity of X: LiH (2 pts)

Team Number: ______

PROBLEM 3
11% of total

TOTAL	A	B	C	D	E	F	G	H
18	**2**	**2**	**2**	**2**	**2**	**4**	**2**	**2**

Write down the net ionic equation for each question below (*does not need to be balanced)*. Also classify the reactions as being combustion, acid-base, organic, decomposition, combination, precipitation, nuclear, oxidation-reduction, or complex-ion formation. Multiple reaction types may apply, but you need only write one.

0.5 pts reactant, 1 pt product, 0.5 pts reaction type

a. Solutions of sodium iodide and lead nitrate are mixed.

$Pb^{2+} + I^{-} \rightarrow PbI_2$

Combination, precipitation

b. Carbon dioxide is bubbled through a solution of calcium hydroxide.

$CO_2 + Ca^{+2} + OH^{-} \rightarrow CaCO_3 + H_2O$

Acid-base, precipitation

c. A solution of hydrofluoric acid is added to a suspension of aluminum hydroxide.

$HF + Al(OH)_3 \rightarrow Al^{+3} + F^{-} + H_2O$

Acid-base

d. Phosphorus is burned in excess oxygen.

$P_4 + O_2 \rightarrow P_4O_{10}$

Combustion, Oxidation-reduction

e. Uranium-238 undergoes alpha decay.

${}^{238}U \rightarrow {}^{234}Th + {}^{4}_{2}\alpha$

Nuclear

f. $YBa_2Cu_3O_7$ is dissolved in aqueous HCl with excess KI in an oxygen-free atmosphere.

$YBa_2Cu_3O_7 + H^{+} + I^{-} \rightarrow Y^{3+} + Ba^{2+} + CuI + H_2O + I_3^{-}$ or

$YBa_2Cu_3O_7 + H^{+} + I^{-} \rightarrow Y^{3+} + Ba^{2+} + CuI + H_2O + I_2$

Acid-base, precipitation, oxidation-reduction (points doubled)

g. Concentrated hydrochloric acid is added to a solution of cobalt(II) chloride and heated.

$Co^{+2} + Cl^{-} \rightarrow CoCl_4^{2-}$

Complex-ion formation

h. Acetic acid is heated with ethanol in the presence of H^{+}

$CH_3COOH + CH_3CH_2OH \rightarrow CH_3COOCH_2CH_3 + H_2O$

Organic

Team Number: ______

PROBLEM 4
7% of total

TOTAL	A	B	C	D	E
12	**3**	**3**	**2**	**2**	**2**

Breathalyzers make an estimate of blood alcohol by measuring the amount of alcohol in a person's breath. Old breathalyzer models functioned by having a person breath through a solution of potassium dichromate, which oxidizes the ethanol in the person's breath into acetic acid. The dichromate changes color in the process, which is measured by a detector.

a. Balance each of the equations below for the oxidation of ethanol and reduction of dichromate.

1 CH_3CH_2OH + 1 $H_2O \rightarrow$ 1 CH_3COOH + 4 H^+ + 4 e^-

1 $Cr_2O_7^{2-}$ + 14 H^+ + 6 $e^- \rightarrow$ 2 Cr^{3+} + 7 H_2O (1.5 pts per correct equation)

b. Combine the two equations to write a net ionic equation.

Equation: *3 CH_3CH_2OH + 3 H_2O + 16 H^+ + 2 $Cr_2O_7^{2-}$ $\rightarrow$ 3 CH_3COOH + 4 Cr^{+3} + 14H_2O*

1 pt reactants, 1 pt products, 1 pt balancing

The concentration of ethanol in the person's blood to ethanol in the person's breath is on average 2100. Thus, this is the standard most machines are calibrated to. The current legal limit on driving in the United States is 0.08g ethanol per 1000cm^3 of blood.

c. If a person at the limit exhales 1L of breath, how many moles of ethanol are breathed out?

Moles of Ethanol = 8.28×10^{-7} mol (2 pts)

d. If 14% of the dichromate molecules in the breathalyzer have reacted, how many molecules of dichromate does the breathalyzer in total contain?

Molecules of Dichromate = 3.324×10^{17} molecules (2 pts)

e. A person's ratio of ethanol in blood to ethanol in breath is 2300. The BAC reported by the breathalyzer will be: *Circle the correct choice*

Higher than the real BAC   Lower than the real BAC (2 pts)

Chemical Reactions & Stoichiometry
1. B
2. C
3. C
4. B
5. A
6. C
7. A
8. B
9. A
10. B

Extended Response: CHEMICAL REACTIONS

a.

i. (+1 point) $2OH^- + SO_3^{2-} \rightarrow SO_4^{2-} + H_2O + 2e^-$

(+1 point) $3e^- + 2H_2O + MnO_4^- \rightarrow MnO_2 + 4OH^-$

ii. (+0.5 point per coefficient, +1 point for whole equation correct)

$2KMnO_4 + 3Na_2SO_3 + H_2O \rightarrow 2MnO_2 + 3Na_2SO_4 + 2KOH$

iii. (+1 point) finding initial moles of reactants as 0.1455 mol $KMnO_4$, 0.1348 mol Na_2SO_3
(+1 point, can be implied) designating Na_2SO_3 as limiting reactant for next calculation
(+1 point) dimensional analysis including 2:3 mol ratio
(+1 point) final answer = 5.045 g KOH

iv. for each, (+1 point) for how, and (+1 point) for correct justification
a) toward reactants / left; removing reactants shifts equilibrium toward reactants due to Le Chatelier's principle (also accept comparison $Q < K$)
b) no change; H_2O is liquid and not included in K or Q expression, and all relevant compounds are soluble, so amount of H_2O does not shift or affect equilibrium
c) toward reactants / left; adding $MgSO_4$ causes concentration of SO_4^{2-} ions

to increase, therefore making $Q < K$ (common ion effect)

b. for each, (+0.5 points) for unbalanced chemical reaction, (+0.5 points) for state symbols, (+0.5 points) for balancing reaction, (+0.5 points) for identifying correct reaction type
i. $Mg(s) + O_2(g) \rightarrow 2MgO(s)$, combustion
ii. $3AgNO_3(aq) + AlCl_3(aq) \rightarrow 3AgCl(s) + Al(NO_3)_3(aq)$, double-replacement
iii. $KHCO_3(s) + HC_2H_3O_2(aq) \rightarrow KC_2H_3O_2(aq) + H_2CO_3(aq)$, double-replacement (also accept acid-base)
iv. $HCl(g) + H_2O(l) \rightarrow H_3O^+(aq) + Cl^-(aq)$, protonolysis (also accept acid-base)